Sahasra Ranjan

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Research & engineering focus

Applied AI research and systems engineering across RL and synthetic environments for agent training and evaluation, eval methodology, real-time multimodal agents, and agent infrastructure. Published research in NLP (ACL 2023 Outstanding Paper Award) and computational biology (Nature Communications 2023); currently building production systems on real-time audio and vision models, and evaluation infrastructure for enterprise AI agents.

Industry experience

Socratic AI Inc Aug 2025 - Present
Founding AI Engineer (2nd engineer) *Sunnyvale, CA*

- Designed and built a synthetic-company simulation environment for **long-horizon evaluation of AI agents**, in which **21 simulated employees** generate email, chat, calendar, and spreadsheet activity calibrated to production telemetry; agents are scored through checkpoint probes and LLM-based graders.
- Developed evaluation infrastructure for real-time and workflow agents, including an LLM-as-judge and prompt-injection test harness, a tool-calling and latency benchmark across real-time models, and regression replay over **3,300+ captured production traces**.
- Architected a unified **real-time speech-to-speech inference layer** across four model providers, handling streaming audio, turn-taking, voice activity detection, and barge-in; supported calls longer than Gemini Live's 15-minute session limit through context compression and session rollover.
- Built a **persistent memory system** for AI agents, comprising an append-only transcript store, source-grounded fact extraction, and two-stage nightly consolidation with rollback safeguards; implemented cost-aware model routing under per-user and organization-wide spend limits.
- Led engineering, reporting to the founder, for the deployment at a publicly traded enterprise communications provider, one of the company's two **multi-year, multi-million-dollar contracts** (the other with a Fortune 150 semiconductor company); introduced provenance validation for AI-generated requirements documents.
- Automated enterprise contact-center migrations through authenticated extraction of 13 configuration categories across region-specific APIs; shipped real-time SDKs in Python, TypeScript, and Swift and an ERP connector that **reduced migrations from six weeks to days**.

Rubrik Inc Summer 2022 (Intern), 2023 - 2024 (Full-time)
Software Engineer, Data Life Cycle Team *Bengaluru, India*

- Designed garbage collection for snapshot metadata, including reference tracking, a mark-and-sweep lifecycle, and gRPC APIs for detecting and removing orphaned entries across distributed storage nodes.
- Extended retention lock and legal hold to cloud-native workloads, enforcing immutability for **petabyte-scale backups**; implemented asynchronous SLA assignment using an event-driven poller across thousands of protected objects.

Entrepreneurship

Mathkraft 2024 - 2025
Co-founder, AI Math Tutoring Platform for K-8 *San Francisco, CA*

- Co-founded an AI tutoring platform that embeds math problems in student-created stories; piloted with **150+ students** at **95%+ retention** over six months.
- Managed a team of 4 engineers and 6 tutors and conducted 50+ in-person tutoring sessions to inform product design.

Technical skills

Languages: Python, C++, Go, TypeScript, Swift, SQL
Machine Learning: PyTorch, JAX, Hugging Face Transformers, vLLM, protein language models
AI Systems: agent evaluation (LLM-as-judge, trace replay), synthetic data generation, agent memory, RAG, model routing
Real-time & Infrastructure: live audio and vision models, WebRTC, SIP, LiveKit, gRPC, Docker, Kubernetes, GCP, PostgreSQL

Selected publications

*359 total citations, h-index 5 (Google Scholar, Oct 2026). *denotes equal contribution.*

1. Richeek Das*, **Sahasra Ranjan***, Shreya Pathak, Preethi Jyothi. *Improving Pretraining Techniques for Code-Switched NLP.* **ACL 2023** Outstanding Paper Award
2. Alexander Kroll, **Sahasra Ranjan**, Martin K. M. Engqvist, Martin J. Lercher. *A general model to predict small molecule substrates of enzymes based on machine and deep learning.* **Nature Communications, 2023** *(265 citations)*
3. Alexander Kroll, **Sahasra Ranjan**, Martin J. Lercher. *A multimodal Transformer network for protein-small molecule interactions enhances predictions of kinase inhibition and enzyme-substrate relationships.* **PLoS Computational Biology, 2024** *(48 citations)*

Research experience

Multimodal Models for Protein-Small Molecule Interactions

Research Intern and Collaborator | Prof. Martin J. Lercher, Alexander Kroll — *HHU Düsseldorf, Germany*

- Developed ESP, a general enzyme-substrate prediction model that jointly encodes protein sequences (fine-tuned ESM-1b) and molecular fingerprints; curated 70k BRENDA pairs with negative sampling of plausible non-substrates, and end-to-end fine-tuning raised accuracy from **82% to 91%**; deployed as a public web server.
- Designed ProSmith, a multimodal transformer with bidirectional cross-attention between protein and molecule representations and contrastive pre-training on 500k BindingDB pairs; reached **94.2% accuracy** on kinase inhibition and enzyme-substrate benchmarks (30% error reduction over the prior state of the art), serving **50k+ predictions** at deepmolecules.org.
- Developed and delivered a substrate-prediction model to a **global pharmaceutical partner**, predicting which drug-metabolizing enzymes and transporters interact with unseen drug molecules; extended a contrastive protein-molecule model (ESM-2, MolFormer) into a six-model ensemble with gradient boosting, reaching **MCC 0.53** and **AUC 0.88** on molecule-held-out splits.
- Designed the evaluation methodology, including molecule-held-out and leakage-controlled (DataSAIL) splits, fixed-ratio negative sampling, and metrics restricted to experimentally confirmed interactions; data audits identified conflicting labels and inflated benchmark performance in public CYP datasets.
- Added uncertainty estimation that reduced expected calibration error from **0.123 to 0.020** and accepts 40% of predictions at an 85% accuracy target; prototyped an LLM-agent pipeline for extracting enzyme-substrate interactions from full-text literature.

Pretraining for Code-Switched NLP — May 2022 - Dec 2022

B.Tech Thesis | Advisor: Prof. Preethi Jyothi, CSALT Lab — *IIT Bombay*

- Showed that standard masked language modeling ignores language boundaries, leading multilingual models (mBERT, XLM-R) to learn spurious monolingual correlations on code-switched text.
- Designed boundary-aware masking strategies, **Switch-MLM** and the label-free **Freq-MLM**, that mask tokens at language switch-points; improved NER and sentiment analysis by **5-7 F1** across Hindi-English, Spanish-English, Tamil-English, and Malayalam-English benchmarks.

Gradient-based Optimization for Continual Learning — Jan 2023 - May 2023

B.Tech Thesis | Advisors: Prof. Preethi Jyothi, Prof. Manoj Prabhakaran — *IIT Bombay*

- Developed **CommitteeSGD**, a continual-learning method that maintains a periodically refreshed buffer of prior examples and modifies current gradients to penalize updates that conflict with learned knowledge, using the local elasticity of neural networks to limit decision-boundary drift; improved accuracy by **3.5%** over standard SGD on CIFAR-100 (VGG16).

Education

Indian Institute of Technology Bombay — Mumbai, India

B.Tech in Computer Science and Engineering | GPA: **9.05**/10 | Honors GPA: **9.75**/10 — *Aug 2019 - May 2023*

Selected honors

- **ACL 2023 Outstanding Paper Award**, co-first author; awarded to less than 1% of accepted papers *2023*
- **Research Excellence Award**, best Bachelor's Thesis among graduating CS students, IIT Bombay *2023*

Teaching

Teaching Assistant, CS725: Foundations of Machine Learning (Prof. Preethi Jyothi), IIT Bombay *Fall 2022*

Teaching Assistant, CS753: Automatic Speech Recognition (Prof. Preethi Jyothi), IIT Bombay *Spring 2023*